KGR-MAS: Knowledge Graph-Enhanced Multi-Agent LLM Framework for Evidence-Based Medical Diagnosis and Clinical Decision Support

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***Abstract*— Large Language Models (LLMs) demonstrate significant potential in healthcare applications, yet they exhibit reasoning bias, information bottlenecks, and limited adaptability in complex clinical settings. To address these limitations, we propose KGR-MAS, which introduces an integrated control loop that couples (1) KG-derived diagnostic paths for executable subtask decomposition and dynamic agent allocation, while using these KG reasoning paths to initialize shared short-term memory; (2) a language-level LSTM-inspired gating mechanism with Transformer-XL short-term memory to regulate cross-agent information flow and maintain global coherence; and (3) a KG-guided RAG-STM process that triggers evidence retrieval at diagnostic checkpoints and writes the retrieved evidence back into STM for subsequent reasoning. We constructed a large-scale medical knowledge graph achieving the best embedding performance (Hits@1:32.8%, MRR:0.498). Experimental results on MedQA and MedMCQA show that KGR-MAS achieves 87.2% accuracy on MedQA(↑14.4%), and 77.4% average accuracy across four open-domain datasets (↑29.0%), outperforming larger models like GPT-o1(72.0%). KGR-MAS provides interpretable, accurate, and clinically grounded diagnostic reasoning, highlighting its potential as a next-generation AI assistant for medical decision-making.**

***Index Terms*— Large Language Models, Multi-agent Systems,Knowledge Graphs, Clinical Decision Support.**

APPENDIX I
KG EMBEDDING AND TRAINING

To optimize knowledge representation for multi-hop relationship reasoning and diagnostic path prediction, we evaluated four state-of-the-art knowledge graph embedding models: TransE, ComplEx, ConvE, and RotatE. Each model was selected for its complementary strengths in capturing different types of medical relationships, and trained using positive and negative samples with appropriate loss functions.

TransE models relationships as translations in the embedding space. The score of a triple $\mathcal{R}_i(\vec{h},\vec{r},\vec{t})$ is defined as the negative L1 norm distance(Eq. (1)), with negative samples created by randomly replacing $\vec{h}$ or $\vec{t}$. The optimization uses Margin Ranking Loss(Eq. (2))

$$f_r(\vec{h},\vec{t}) = -\|\vec{h}+\vec{r}-vect\|_p \tag{1}$$

$$L = \sum_{(\vec{h},\vec{r},\vec{t})\in\mathcal{T}} \sum_{(h',r,t')\in\mathcal{N}_{(h,r,t)}} \max\Big(0,\ \gamma + f_r(\vec{h},\vec{t}) - f_r(\vec{h'},\vec{t'})\Big) \tag{2}$$

ComplEx extends the idea of DistMult and introduces complex-valued vectors to capture asymmetric relationships. It can represent both symmetric and asymmetric relations and is suitable for a variety of relational patterns. The scoring function is defined in Eq. (3), and the model is trained using binary cross-entropy (BCE) loss (Eq. (4)). After scoring, the result is mapped to the interval $[0,1]$ through the sigmoid function $\sigma(\cdot)$.

$$f_r(\vec{h},\vec{t}) = \mathrm{Re}\Big(\langle\vec{h},\ \vec{r},\ \overline{\vec{t}}\rangle\Big) = \mathrm{Re}\Big(\sum_{k=1}^{d} h_k\cdot r_k\cdot\overline{t_k}\Big). \tag{3}$$

$$L = -\sum_{pos}\log\sigma\big(f_r(\vec{h},\vec{t})\big) \ - \sum_{neg}\log\sigma\big(-f_r(\vec{h'},\vec{t'})\big). \tag{4}$$

ConvE rearranges the head entity vector $\vec{h}$ and the relationship $\vec{r}$ into two dimensions, performs 2D convolution in the convolutional network, and then projects the output score. ConvE uses convolution filtering to extract the spatial features of the combination of $\vec{h}$ and the relationship $\vec{r}$, so as to better model complex relationships, using cross entropy (BCE) loss(Eq. (5),Eq. (4)).

$$f_r(h,t) = \Big(\mathrm{vec}\big(\mathrm{Conv2D}([\vec{h};\vec{r}])\big)\cdot W\Big)\cdot\vec{t}. \tag{5}$$

RotatE is an embedding method based on complex space rotation transformation. It uses distance metric (Eq. (6)) for

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scoring. Similar to TransE, it can also use margin ranking loss function(Eq. (7),Eq. (8)).

$$d(h,r,t) = \|\vec{h} \circ \vec{r} - \vec{t}\| \tag{6}$$

$$f_r(\vec{h},\vec{t}) = -\,d(\vec{h},\vec{r},\vec{t}) \tag{7}$$

$$L = \sum_{(\vec{h},\vec{r},\vec{t})\in pos} \sum_{(\vec{h'},\vec{r},\vec{t'})\in neg} \max\Big(0,\ \gamma + d(\vec{h},\vec{r},\vec{t}) - d(\vec{h'},\vec{r},\vec{t'})\Big). \tag{8}$$

APPENDIX II
LSTM-INSPIRED GATING AND SUBGRAPH ANALYSIS

Appendix II provides additional technical details omitted from the main text for brevity. Subsection A describes the full parameterisation of the LSTM-inspired gating module, including the pre-trained checkpoint, the sensitivity analysis of α, and the training objective and optimisation protocol for future end-to-end fine-tuning. Subsection B then presents quantitative evidence that diagnostic subgraph extraction effectively refocuses the knowledge graph toward symptom–disease reasoning.

A. Parameterisation and Training Objective of the Gating Mechanism

1) Parameter Categories: The LSTM-inspired gating module described in Section .C contains three distinct categories of parameters. We describe each in detail below.

Category 1: Pre-trained Transformer-XL attention weights.

The short-term memory(STM) update rule in Eq. (13), is implemented using the publicly available `transfo-xl-wt103` checkpoint. This model was pre-trained on the WikiText-103 corpus and provides robust long-range sequence modelling with segment-level recurrence and relative positional encodings, which are well-suited to the sequential, multi-agent information flow in KGR-MAS.

The attention projection matrices $\mathbf{W}_Q$, $\mathbf{W}_K$, $\mathbf{W}_V$ of this checkpoint serve as the concrete instantiation of the abstract weight matrices $\mathbf{W}^{(F)}$ and $\mathbf{W}^{(I)}$ appearing in the forget gate and input gate definitions:

$$G_i^{(F)} = \sigma\Big(\mathbf{W}^{(F)} \cdot \mathrm{CarryOver}_{i-1} + \mathbf{b}^{(F)}\Big), \tag{11}$$

$$G_i^{(I)} = \sigma\Big(\mathbf{W}^{(I)} \cdot [\mathrm{Output}_i;\ \mathrm{Prompt}_i] + \mathbf{b}^{(I)}\Big). \tag{12}$$

In the current KGR-MAS deployment, these weights are frozen during inference: no gradient-based update is applied to $\mathbf{W}^{(F)}$, $\mathbf{W}^{(I)}$, $\mathbf{b}^{(F)}$, or $\mathbf{b}^{(I)}$ during the experiments reported in Section III. This design choice is deliberate. Annotated corpora of complete multi-step diagnostic reasoning trajectories with ground-truth intermediate states do not yet exist at the scale required for reliable fine-tuning. Reusing a high-quality pre-trained sequence model avoids overfitting to the limited available supervision while still providing the long-range dependency modelling that the cross-agent STM update requires.

The forget gate activation f_i shown in Fig.2 corresponds to $G_i^{(F)}$ in Eq. (11); the symbol o_i corresponds to the input-update gate $G_i^{(I)}$ in Eq. (12), which controls the proportion of the current agent output written into STM.

Category 2: The mixing coefficient α.

The CarryOver update rule(Eq.(10)) contains the scalar $\alpha \in (0,1)$, which controls the relative weight given to the most recent agent output versus the accumulated prior context. This parameter is not gradient-learned; it is treated as a tunable hyper-parameter.

To determine the optimal value, we performed a grid search over $\alpha \in \{0.1, 0.2, 0.3, 0.4, 0.5\}$ using 100 held-out development cases drawn from the MedDX-OOD dataset via stratified random sampling (same sampling protocol as the main evaluation in Section III.E). Table IX reports the resulting validation accuracies. These 100 development cases were sampled separately from and prior to the 200 test cases used in Section III.E, ensuring no overlap between hyperparameter tuning and final evaluation.

The plateau around $\alpha = 0.3$ reflects a meaningful balance: a value that is too small ($\alpha \le 0.1$) causes the system to over-rely on stale accumulated context and fail to incorporate new agent findings promptly, while a value that is too large ($\alpha \ge 0.5$) causes the system to over-weight the most recent agent output and discard valuable prior evidence. $\alpha = 0.3$ was therefore adopted as the default for all experiments.

TABLE I: Validation accuracy (%) under different values of the mixing coefficient α.

α	0.1	0.2	0.3	0.4	0.5
Val. Acc. (%)	68.5	73.2	**79.3**	75.8	70.1

Note: Results are reported on 100 MedDX-OOD development cases, sampled independently from and prior to the 200 test cases used in Section III.E, ensuring no overlap between hyperparameter selection and final evaluation. We set $\alpha = 0.3$ in all experiments reported in the main paper.

Category 3: Evidence score magnitudes.

The evidence scores $s_i^{(j)} \in [-1,+1]$ in Section II.D are assigned by each specialised agent based on compatibility between clinical findings and disease characteristics. These are not learned parameters; they are outputs of the LLM backbone's natural language inference capability, calibrated by the structured prompt template provided to each agent. The threshold boundaries ($|s| > 0.6$: strong; $0.3 < |s| \le 0.6$: moderate; etc.) were defined based on standard clinical evidence-grading conventions (analogous to the GRADE framework) and were held constant across all experiments.

2) Complete Training Objective for End-to-End Fine-Tuning: For full reproducibility of any future training run in which all parameters ($\mathbf{W}^{(F)}$, $\mathbf{W}^{(I)}$, $\mathbf{b}^{(F)}$, $\mathbf{b}^{(I)}$, and α) are learned jointly, we provide a self-contained specification below.

Let $\mathcal{D}_{\text{train}} = \{(\mathbf{x}_n, y_n)\}_{n=1}^{N}$ denote a labelled training set of N diagnostic cases, where $\mathbf{x}_n$ is the full patient narrative (free

text) and $y_n \in \{d_1, d_2, \ldots, d_m\}$ is the confirmed ground-truth diagnosis. Each case induces a reasoning trajectory of T agent steps, producing intermediate STM states $\mathrm{STM}_1, \ldots, \mathrm{STM}_T$ and a final diagnosis probability distribution $P_{\text{final}}(\cdot)$ via Eq.(21).

The joint training objective is:

$$\mathcal{L} = \frac{1}{N} \sum_{n=1}^{N} \Big[\mathcal{L}_{\text{CE}}(\hat{y}_n,\, y_n) \;+\; \lambda\, \mathcal{L}^{(n)}_{\text{consistency}} \Big], \tag{9}$$

where the two terms are defined as follows.

Diagnostic cross-entropy loss:

$$\begin{aligned} \mathcal{L}_{\text{CE}}(\hat{y}_n, y_n) &= -\log P_{\text{final}}(y_n) \\ &= -\log \frac{\exp\!\big(\log P_0(y_n) + \sum_{i=1}^{T} s_i^{(y_n)}\big)}{\sum_{k=1}^{m} \exp\!\big(\log P_0(d_k) + \sum_{i=1}^{T} s_i^{(k)}\big)}. \end{aligned} \tag{10}$$

where $P_0(d_k)$ is the initial KG-derived diagnosis probability from Eq. (3), $s_i^{(k)}$ is the evidence score assigned to diagnosis d_k at step i (Section II.D), and P_{final} is the normalised product given by Eq.(21). This loss directly optimises the end-to-end diagnostic accuracy of the full multi-agent pipeline.

STM consistency regulariser:

$$\mathcal{L}^{(n)}_{\text{consistency}} = \frac{1}{T} \sum_{t=1}^{T} \big\| \mathrm{STM}_t^{(n)} - \mathrm{STM}_{t-1}^{(n)} \big\|_2^2, \tag{11}$$

where $\mathrm{STM}_t^{(n)}$ denotes the STM state after agent step t for case n. This term penalises abrupt, large-magnitude transitions in the shared STM, encouraging the gating mechanism to produce smooth, coherent memory updates rather than volatile state changes that could corrupt the global diagnostic context accumulated across agents. The two objectives are complementary: $\mathcal{L}_{\text{CE}}$ drives overall diagnostic correctness, while $\mathcal{L}_{\text{consistency}}$ ensures the quality of the intermediate reasoning chain.

The regularisation coefficient λ is set to 0.01 by default. Its sensitivity can be assessed via grid search over $\lambda \in \{0.001, 0.01, 0.1\}$ on a held-out validation set; in pilot experiments we found that $\lambda = 0.01$ consistently dominated across all tested datasets.

- **Learning rate:** $\eta = 1 \times 10^{-5}$, one order of magnitude lower than the KG embedding learning rate (1×10^{-4}) to prevent catastrophic forgetting of the pre-trained Transformer-XL representations during fine-tuning.
- **Schedule:** Cosine annealing with warm-up over the first 10% of training steps.
- **Batch size:** 16 cases per gradient step.
- **Maximum epochs:** 50, with early stopping on validation accuracy (patience = 5 epochs).
- **Gradient clipping:** Maximum ℓ_2 norm of 1.0, applied to all trainable parameters.
- **Hardware:** Training can be completed on a single NVIDIA A100 (80 GB) GPU. The computational cost scales with training corpus size N and the number of agent steps T per case.

The only required supervision is the confirmed ground-truth diagnosis label y_n at the case level. No intermediate supervision on individual agent outputs, individual STM states, or individual evidence scores is required. This makes the training objective compatible with standard clinical datasets that record only final diagnoses without requiring manual annotation of reasoning steps.

As described in Section II.C, the experiments reported in Section III use frozen Transformer-XL weights and a fixed $\alpha = 0.3$. The training objective specified above is provided for reproducibility of future fine-tuning experiments. End-to-end fine-tuning requires a labelled corpus of confirmed diagnostic cases with complete patient narratives; constructing such a corpus with appropriate ethical oversight and privacy-preserving protocols is identified as an important direction for future work.

B. Knowledge Graph Diagnostic Subgraph Relation Distribution

1) Motivation: As noted in Section III.B, the full KG contains 81.19% Drug–Drug relations (Table I), while the diagnostic reasoning module focuses on symptom–disease subgraphs. This section provides quantitative evidence that the subgraph extraction algorithm structurally resolves this apparent mismatch at inference time, and explains why the embedding model trained on the full KG nonetheless produces high-quality representations for the minority disease-symptom relation type.

2) Subgraph Extraction Protocol: For each patient case, the Task Planner Agent (TPA) extracts a diagnostic subgraph $\mathcal{G}_s^{(k)}$ from the full KG via breadth-first traversal starting from all identified symptom nodes, up to depth $k \in \{2, 3, 4\}$ hops. The depth k is selected dynamically based on the number of confirmed symptom nodes: The depth $k \in \{2, 3, 4\}$ is selected to balance diagnostic coverage and computational efficiency, consistent with the maximum subgraph size of 200 nodes and mean generation time of 2.74 ± 0.31s reported in Section III.C.

The traversal is *symptom-anchored*: only nodes reachable from at least one confirmed symptom node within k hops are included in $\mathcal{G}_s^{(k)}$. Drug–Drug edges connect drug nodes to other drug nodes; since drug nodes are not symptom nodes and are not directly connected to symptom nodes in the graph topology of our KG, they are not reachable from symptom anchors in $k \leq 4$ hops in the vast majority of cases. This structural property of the graph—not an explicit filtering rule—is what suppresses Drug–Drug edges in the diagnostic subgraphs.

3) Quantitative Relation Distribution Analysis: To empirically verify this suppression, we extracted the diagnostic subgraph for each of the 500 retrospective clinical cases used in Section III.C and computed the proportion of each relation type among all edges in each subgraph. Table X reports the mean and standard deviation across the 500 cases.

The results indicate that subgraph extraction effectively redirects the original KG toward diagnosis-relevant relations. Drug–Drug edges are reduced from 81.19% in the full KG to $6.8\% \pm 3.5\%$ in the extracted subgraphs, while Disease–Phenotype edges increase from 9.20% to $64.8\% \pm 7.6\%$, becoming the dominant relation type. This shift aligns well with

TABLE II: Relation-type distribution in the full KG and the extracted diagnostic subgraphs.

Relation Type	Full KG (%)	Diagnostic Subgraph (%)
Disease–Phenotype	9.20	$\mathbf{64.8 \pm 7.6}$
Disease–Disease	1.96	13.1 ± 3.4
Phenotype–Phenotype	1.14	8.3 ± 2.9
Drug–Effect	3.94	5.2 ± 2.1
Indication / Contraindication	2.43	1.2 ± 0.8
Exposure–Disease	0.14	0.6 ± 0.4
Drug–Drug	81.19	6.8 ± 3.5
Total	**100.00**	**100.0**

Note: Diagnostic subgraphs $\mathcal{G}_s^{(k)}$ were extracted during inference across all 500 retrospective clinical cases. Values for diagnostic subgraphs are reported as mean $\pm$ standard deviation across cases. Disease–Phenotype combines Disease_Positive_Phenotype (9.13%) and Disease_Negative_Phenotype (0.07%) from Table I. Indication / Contraindication combines Indication (0.57%) and Contraindication (1.86%).

the requirements of differential diagnosis, where symptom–disease associations play a central role. The remaining relation types, including Disease–Disease, Drug–Effect, and Exposure–Disease, also retain clear clinical relevance by supporting comorbidity reasoning, treatment-related interpretation, and environmental risk assessment. These findings suggest that the extracted subgraphs are both more focused and diagnostically meaningful than the full KG.

4) Embedding Robustness to Distributional Skew: A separate but related question is whether the embedding model trained on the full KG (81.19% Drug–Drug) produces high-quality vector representations specifically for the minority disease-symptom relation type. Two training-design choices directly address this concern.

During RotatE training, negative samples are drawn using frequency-weighted sampling, which assigns higher probability to negatives involving entity types that occur less frequently as heads or tails in the training set. Since Disease and Symptom nodes are much less frequent than Drug nodes, which dominate the KG through the 81.19% Drug–Drug triples, the sampler correspondingly up-weights negatives associated with Disease–Symptom triples. This provides sustained gradient signal for disease-symptom relational patterns and helps prevent the model from overfitting to the dominant Drug–Drug structure.

Unlike standard LCWA, which treats all unobserved triples as negative, sLCWA samples only a subset of unobserved triples as negatives at each training step. This is especially important for sparse relations such as Disease–Symptom and Exposure–Disease. Under LCWA, many unobserved but potentially valid triples would be incorrectly treated as negative, introducing false-negative noise and biasing the model away from clinically important associations. By sampling negatives stochastically, sLCWA reduces this contamination and lowers the chance that a true but unobserved triple is repeatedly used as a negative example.

The benefit of these design choices is further reflected in the downstream Symptom–Disease Association task (Table IV), where RotatE achieves 91.7% Precision@3 on 5,000 curated symptom–disease pairs. Despite the pronounced relational imbalance of the training corpus, this result suggests that the learned embeddings capture clinically meaningful diagnostic associations rather than merely fitting the dominant Drug–Drug structure. In comparison, ComplEx reaches 84.1% Precision@3, while ConvE achieves only 52.8% despite near-zero training loss (Fig. 3). These results indicate that training-set fit alone is insufficient, and that RotatE's advantage arises from its stronger inductive bias for modeling medically relevant relational structure.